

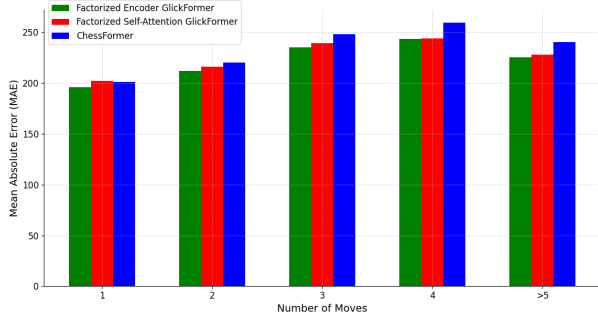


Fig. 4: Mean Absolute Error of each model depending on number of moves required to solve the puzzle.

The standard distribution sampling was sufficient to prevent the models from overfitting, eliminating the need for additional regularization techniques such as dropout or weight decay.

B. Baseline Model

To evaluate the effectiveness of our proposed models, we decided to compare it against a state-of-the-art solution. Therefore we have chosen a baseline model based on ChessFormer architecture. The model processes sequences of board states by stacking them along the input channels, following ChessFormer’s method of incorporating temporal information by stacking the past N boards. This approach embeds temporal information directly into the input before it is processed by the transformer encoder. Furthermore, we utilize ChessFormer’s value head to process the features extracted from the encoder in the same way as in our GlickFormer models. By comparing against this baseline, we can assess the impact of our proposed methods for modelling temporal dynamics on the performance of puzzle difficulty prediction.

C. Results and Analysis

We define three metrics to evaluate the performance of models on the test set. **Mean Absolute Error (MAE)**, **Mean Absolute Z-Score (MAZ)**, and **Accuracy within 1, 2, and 3 standard deviations**.

1. **Mean Absolute Error (MAE):** The *Mean Absolute Error* is given by the formula:

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^N |r_i - \hat{r}_i|$$

where r_i represents the ground truth Glicko-2 ratings, $\hat{r}_i$ are the predictions, and N is the total number of test samples. The MAE measures the average magnitude of errors in the predictions, indicating how far off the predictions are from the actual values on average.

2. **Mean Absolute Z-Score (MAZ):** The *Mean Absolute Z-Score (MAZ)* is calculated by normalizing the absolute error using the standard deviation of the ground truth values:

$$\text{MAZ} = \frac{1}{N} \sum_{i=1}^N \left| \frac{r_i - \hat{r}_i}{RD_i} \right|$$

where RD_i is the rating deviation associated with the ground truth rating value r_i . The MAZ adjusts for the uncertainty of the ratings, allowing us to compare the magnitude of errors relative to the expected variation in the ground truth.

3. Accuracy within 1, 2, and 3 Rating Deviations:

This metric measures the proportion of predictions that fall within 1, 2, and 3 standard deviations of the true values. In the context of the Glicko-2 rating system, the rating deviation (RD) reflects the uncertainty in a puzzle’s difficulty rating, analogous to the standard deviation (σ) in a normal distribution.

For normally distributed data, the empirical rule states the following probabilities:

- **68%** of the data lies within 1 standard deviation (σ) of the mean.
- **95%** of the data lies within 2 standard deviations (2σ).
- **99.7%** of the data lies within 3 standard deviations (3σ).

This rule remains valid for the Glicko-2 system. For example, a player with a rating of 1500 and an RD of 50 has a real strength between 1400 and 1600 (two standard deviations from 1500) with 95% confidence. Similarly, for puzzle difficulty ratings, the RD provides a confidence interval around the estimated difficulty.

The accuracy for each case is computed as:

$$\text{Accuracy within } kRD = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(|r_i - \hat{r}_i| \leq kRD_i)$$

where $k = 1, 2, 3$, and $\mathbb{I}(\cdot)$ is the indicator function that returns 1 if the condition inside is true and 0 otherwise. This metric allows us to understand how many predictions fall within an acceptable range of variability from the true values.

TABLE I: Test Set Mean Absolute Error (MAE) and Mean Absolute Z-Score (MAZ)

Model	MAE	MAZ
ChessFormer	227.00	2.68
Factorized Self-Attention GlickFormer	221.80	2.62
Factorized Encoder GlickFormer	217.71	2.57

TABLE II: Test Set Accuracy within 1, 2, and 3 Rating Deviations

Model	Accuracy (%)		
	1RD	2RD	3RD
ChessFormer	25.42	48.21	65.29
Factorized Self-Attention GlickFormer	26.13	48.54	66.08
Factorized Encoder GlickFormer	27.00	50.45	67.66

The results of our experiments are presented in Tables I and II. Our proposed models, Factorized Self-Attention GlickFormer and Factorized Encoder GlickFormer, outperform the baseline ChessFormer model in all metrics.

The Mean Absolute Error (MAE) and Mean Absolute Z-Score (MAZ) are lower for our models, indicating more accurate predictions. Specifically, the Factorized Encoder GlickFormer achieves the lowest MAE of 217.71, demonstrating the effectiveness of separating spatial and temporal processing.

The accuracy within 1, 2, and 3 rating deviations also shows improvement with our models. The Factorized Encoder GlickFormer achieves the highest accuracy within all deviation ranges, with 67.66% of predictions within 3 rating deviations of the true ratings.

Figure 4 depicts the distribution of MAE across puzzles in the test set, categorized by the number of moves required to solve them. This visualization highlights the enhanced performance of our models over the baseline, particularly in puzzles requiring multiple moves. However, an exception is observed with single-move puzzles, where the Factorized Self-Attention GlickFormer performed worse than the baseline.

These results confirm that incorporating temporal information significantly enhances the model’s ability to predict puzzle difficulty. Modeling the sequence of moves allows the model to understand how the complexity of the puzzle evolves, leading to more accurate difficulty assessments.

Additionally we report the MSE on the test set of the IEEE BigData 2024 Cup: Predicting Chess Puzzle Difficulty competition. The prototype version of our Factorized Encoder model achieved a preliminary test set MSE score of 75,995 and a final MSE score of 158,292.

VI. CONCLUSION

In this paper, we presented **GlickFormer**, a transformer-based model designed to predict the difficulty of chess puzzles by approximating the Glicko-2 rating system. By effectively capturing both the spatial and temporal dynamics inherent in chess puzzles, GlickFormer demonstrates the potential of transformer architectures in modeling complex human cognitive tasks beyond traditional gameplay.

Our approach leverages the ChessFormer backbone for spatial feature extraction and incorporates temporal information using methods inspired by prior work on factorized transformer architectures, specifically the *Factorized Encoder Model* and the *Factorized Self-Attention Model*. Through extensive experiments on a large dataset of over 4 million chess puzzles, we showed that GlickFormer outperforms the baseline state-of-the-art ChessFormer model across multiple metrics, including Mean Absolute Error, Mean Absolute Z-Score, and accuracy within 1, 2, and 3 rating deviations.

The results indicate that modeling the sequential nature of puzzles significantly enhances the ability to predict their difficulty, highlighting the importance of temporal information in understanding human problem-solving processes. Our models not only provide more accurate difficulty assessments but also demonstrate the applicability of transformer architectures in capturing abstract cognitive challenges, rather than merely optimizing for game outcomes.

This work contributes to the field by offering a new perspective on evaluating chess puzzle difficulty and underscores the potential of advanced deep learning models in educational and training contexts. Future work may include exploring more advanced temporal modeling techniques, such as incorporating attention mechanisms that focus on critical moves within a puzzle. Furthermore, integrating additional

contextual information, such as player engagement data or thematic elements of puzzles, could further refine difficulty predictions. Extending this approach to other domains that require modeling human cognitive processes could provide valuable insights and applications.

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